AWS EC2 Auto Scaling Lab Tasks

Task 1: Create a Launch Template

Create a launch template with Amazon Linux/Ubuntu, t2.micro, SSH and HTTP security group, key pair, and user data to install Apache/Nginx and display the hostname.

Task 2: Create a Target Group

Create an instance target group using HTTP health checks on '/' without manually registering instances.

Task 3: Create an Application Load Balancer

Create an internet-facing ALB in two public subnets and attach the target group.

Task 4: Create an Auto Scaling Group

Use the launch template, attach the ALB target group, choose two public subnets, and configure Min=2, Desired=2, Max=5.

Task 5: Manual Scaling

Increase desired capacity from 2 to 4, verify new instances, then reduce back to 2 and observe instance termination.

Task 6: Dynamic Scaling Policy

Configure a target tracking policy with Average CPU Utilization target of 50%.

Task 7: Generate CPU Load

SSH into an instance, install stress, generate CPU load above 50%, and monitor CloudWatch and Auto Scaling activities.

Task 8: Automatic Scale-In

Stop the stress test and observe the Auto Scaling Group terminate extra instances after CPU utilization drops.

Task 9: Verify Load Balancing

Access the ALB DNS repeatedly and confirm traffic is distributed across different instances.

Task 10: Monitor Scaling Activities

Review Auto Scaling activity history, CloudWatch metrics, target group health, and identify launch/termination events.